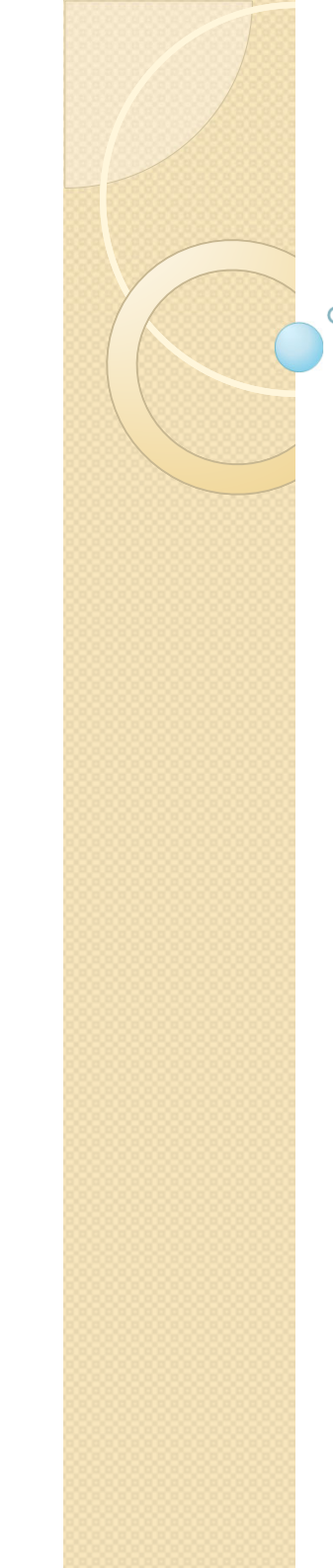




Project Based Learning Course Classes 5 and 6

Overall comments for the presentations of
the conceptual design

System design

2025/10/17

Kouzou Ohara

Outline of Class 5 and 6

- Feedback to the presentations of conceptual design
- System design
 - ✓ Specification design
 - ✓ Structural design
 - ✓ Schedule and role assignment

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Relation between Conceptual Design and System Design

Conceptual Design

Breaking
down



Provide the clear definition of the problem your team will address
(users, technologies, available data, importance, usefulness, etc.)

System Design

Design and systematically organize concrete modules that compose your solution application

✓ Please see the e-learning contents for the details.

Before considering Specification Design

Please confirm your conceptual design again.
Is it clear enough to move ahead your project?

- ✓ Who are stakeholders?
- ✓ Who are users?
- ✓ What is your goal?
 - What is the input to your system?
 - What is the output of your system?
- ✓ What technologies do you need to achieve your goal?
- ✓ Do you have enough data for that purpose?
- ✓ Do there exist missing values?
- ✓ Do there exist outliers?

What you should determine in Specification Design (I)

You should clarify the following items **for the system you will actually implement** :

- ✓ Functions to be implemented
 - ➔ List concrete functions to satisfy users' requirements.
- ✓ System configuration to provide the functions

What components comprise your system?

 - ➔ Make a diagram that presents how the components connect to each other.
- ✓ System sequences with user interface

How do the users use the system?

 - ➔ Make flows of concrete screen images of user interface for each expected usage scenario.

What you should determine in Specification Design (2)

- ✓ Data types to be used

What data does the system actually use?

➔ List concrete relational schemata to be used in your system.

In the e-learning slides, you can find examples of the system configuration, system sequences, and data types.

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- Feedback to the presentations of conceptual design
- **System design**
 - ✓ Specification design
 - ✓ **Structural design**
 - ✓ Schedule and role assignment
- Relationship between this course and the other courses

What you should determine in Structural Design (I)

You should clarify the following items **for the system you will actually implement** :

- ✓ Operating environment of your system
On what OS does your system work?
 - ➔ Show OS and its version. Besides, specify the name of the database system if needed.
 - ✓ Development language
What programming languages are used to implement each component of your system?
 - ➔ Show programming languages for each component.
- * Basically, we assume using Python for AI engine.

What you should determine in Structural Design (2)

- ✓ Tools to be used for development

Which IDE is used? What libraries are required?

➔ Show your development environment and list external libraries you need.

- ✓ Modules to be implemented

What modules are required to implement each function of your system?

How do they communicate to each other?

➔ Break down each function into independent components, diagram their relationships, and define their communications.

What you should determine in Structural Design (3)

- ✓ Communication protocols

If your system is not stand-alone, what protocol is used to communicate between subsystems?

➔ Show protocols for each communication between subsystems, define messages to be communicated, and determine how to deal with errors.

In the e-learning slides, you can find the requirements that modules must satisfy.

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Schedule and role assignment

In the next presentation, you are required to show the development schedule and role assignment.

- ✓ When is the deadline to implement each module?
Who will implement each module?
- ✓ When and who will conduct unit tests for each module?
- ✓ When and who will conduct operation tests?
- ✓ Who will manage the overall schedule?

Making a Gantt chart is a common way to create and manage a development schedule.

As for the unit and operation tests, you should show in detail how your modules and system are tested. See the e-learning slides for the details of these tests.

About the Overall Schedule

- ✓ You must create a schedule that allows you to complete the implementation and testing of the proposed system within the class period.
- ✓ To this end, during system design, clarify the sub-tasks needed to realize your system, who will handle each one, and by when. Some tasks may be carried out in parallel.
- ✓ It is important to anticipate potential difficulties when creating the schedule. Risk management, how to deal with such difficulties, is a critical aspect of real-world projects.



**Let's start thinking
with your group members!**

Use of Data Provided

- ✓ Using the provided data solely for problem detection (observation) is not adequate.
- ✓ Examples of the use of the data
 - Training and/or testing a machine learning model
 - Acquiring the knowledge to guide your solution through detailed analysis
 - e.g.) Through analysis, it may be possible to know what additional data can be related.
- ✓ You can use additional data if necessary.